

**PRAKTIKUM SISTEM OPERASI**

**MODUL 10**

**SHELL**



Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

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**2026**

## JURNAL

1. Akan dimodifikasi shell dengan modifikasi syscall bernama cname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

Perhatikan sekarang syscall cname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h cname diubah menjadi:

```
extern status ascdade(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file cname.c */
extern pri16 cname(pid32, char *, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);
```

- b. Pada cname.c fungsi diubah dari:

```
// Nama: Rizaldy Aulia Rachman
// NIM: 2311104051

#include <xinu.h>

/*-----
 * cname - Dummy syscall
 *-----
 */
syscall cname(
    pid32      pid,
    char *     newname
)
```

menjadi

```
/* chrname.c - chrname */

// Nama: Rizaldy Aulia Rachman
// NIM: 2311104051

#include <xinu.h>

/*-----
 * chrname - Dummy syscall
 *-----
 */
pris chrname(
    pid32      pid,
    char *     newname,
    uint32     namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil

```

/*
 *
 */
pri16 chname(
    pid32 pid,
    char *newname,
    uint32 namelen
)
{
    intmask mask;
    struct proctab *prptr;
    uint32 i;

    mask = disable();

    /* Validasi PID */
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

    /* Nama baru */
    for (i = 0; i < namelen && i < PNMLEN-1; i++) {
        prptr->prname[i] = newname[i];
    }

    /* Null terminator */
    prptr->prname[i] = '\0';

    restore(mask);
    return OK;
}

```

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

a. Edit shell.c

```

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/****
 * Table of Xinu shell commands and the function associated with each
 ****/
const struct cmdent cmdtab[] = {
    {"argecho", xsh_argecho},
    {"arp", xsh_arp},
    {"cat", xsh_cat},
    {"clear", xsh_clear},
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"mемdump", xsh_memdump},
    {"memstat", xsh_memstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep},
    {"tee", xsh_tee},
    {"udp", xsh_udpdump},
    {"udpecho", xsh_udpecho},
    {"udpserver", xsh_udpserver},
    {"uptime", xsh_uptime},
    {"namecmd", xsh_namecmd},
    {"mycmd", xsh_mycmd},
    {"?", xsh_help}
}

```

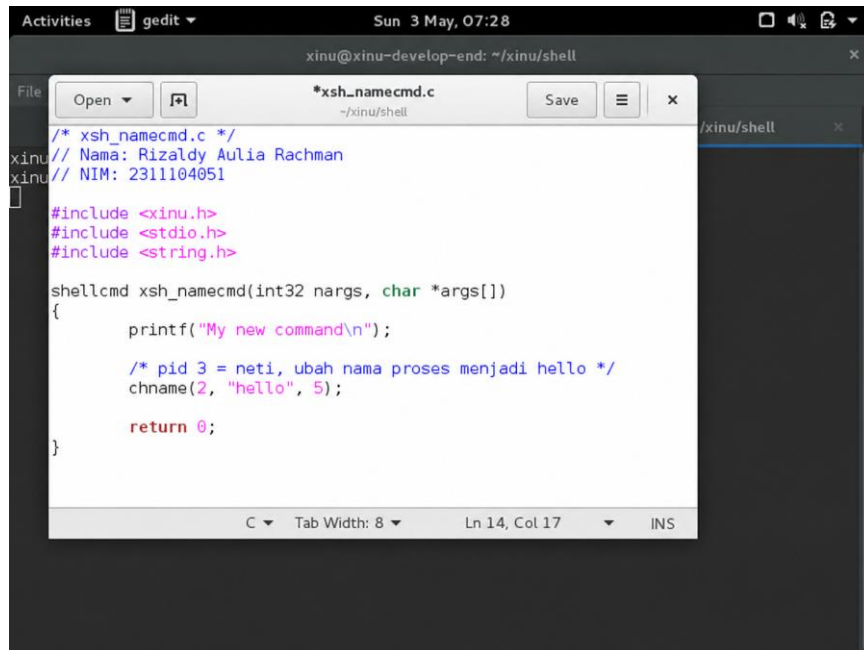
b. Edit shprototypes.h

```

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

```

c. Buat file baru:



```

/* xsh_namecmd.c */
// Nama: Rizaldy Aulia Rachman
// NIM: 2311104051

#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int32 nargs, char *args[])
{
    printf("My new command\n");

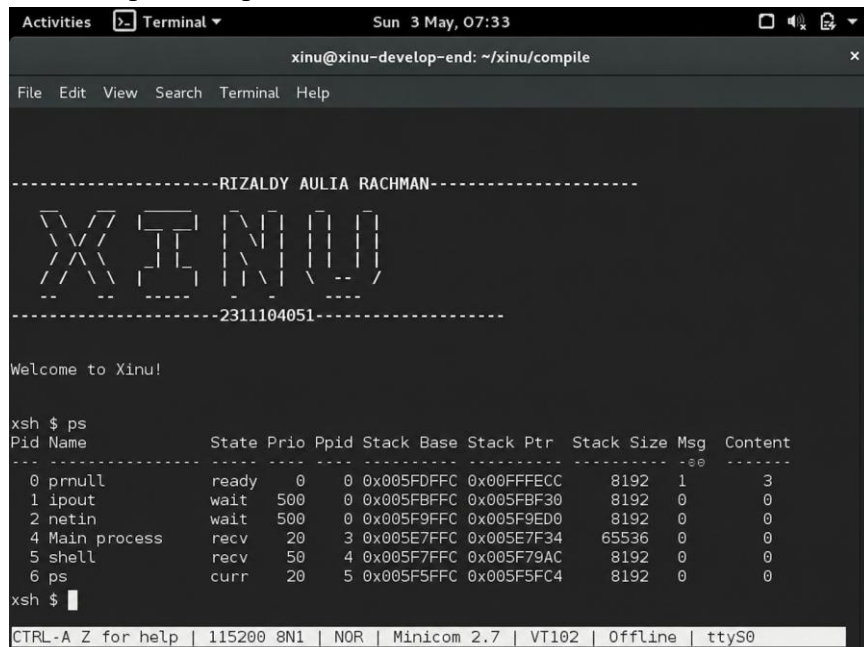
    /* pid 3 = neti, ubah nama proses menjadi hello */
    chname(2, "hello", 5);

    return 0;
}

```

3. Test hasilnya:

- a. Masuk ke terminal xinu
- b. Jalankan perintah ps



```

-----RIZALDY AULIA RACHMAN-----
XINU
-----2311104051-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout            wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin            wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process     recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell            recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps               curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0
xsh $

```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

- c. Jalankan perintah `namecmd`

```

Activities Terminal Sun 3 May, 07:33
xinu@xinu-develop-end: ~/xinu/compile

-----RIZALDY AULIA RACHMAN-----
XINU
-----2311104051-----

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait  500  0  0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0  0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3  0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4  0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr  20   5  0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

- d. Jalankan perintah `ps`

```

Activities Terminal Sun 3 May, 07:33
xinu@xinu-develop-end: ~/xinu/compile

-----2311104051-----

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait  500  0  0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0  0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3  0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4  0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr  20   5  0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE08 8192 1 3
1 ipout       wait  500  0  0x005FBFFC 0x005FBF30 8192 0 0
2 hello       wait  500  0  0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3  0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4  0x005F7FFC 0x005F79AC 8192 0 7
6 ps         curr  20   5  0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

- e. Lihat nama proses telah berubah

Nama proses berubah dari *netin* menjadi *hello* karena syscall `chname()` berhasil dijalankan melalui command `namecmd`.